Exercise J-M

1. If the point on the curve $y^2 = 6x$, nearest to the point $\left(3, \frac{3}{2}\right)$ is (α, β) then $2(\alpha + \beta)$ is equal to _____.

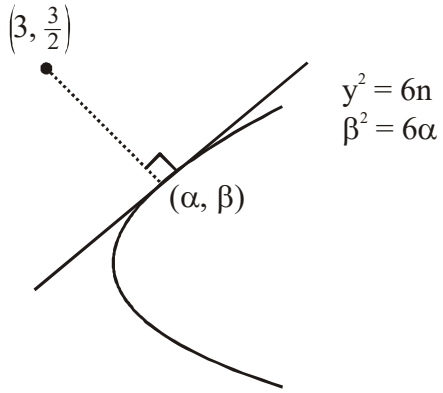
[JEE(Main)July2021]

यदि वक्र $y^2 = 6x$ पर, बिन्दु $\left(3, \frac{3}{2}\right)$ के निकटतम बिन्दु (α, β) है, तो $2(\alpha + \beta)$ बराबर है _____

[JEE(Main)July2021]

Ans. 9

Sol.



$y^2 = 6n$

$2y\frac{dy}{dx} = 6$

$\frac{dy}{dx} = \frac{3}{y} = \frac{3}{\beta}$

$\frac{\beta - \frac{3}{2}}{\alpha - 3} = \frac{-\beta}{3}$

$3\beta - \frac{9}{2} = -\alpha b + 3\beta$

$\alpha\beta = \frac{9}{2}$

$\beta^2 = 6\alpha$

$\beta = \frac{9}{2\alpha}$

$6\alpha = \frac{81}{4\alpha^2}$

$\alpha^3 = \frac{27}{8} \quad \Rightarrow \quad \alpha = \frac{3}{2}$

$\beta = \pm 3$

$\alpha = \frac{3}{2} \quad \beta = 3$

$2(\alpha + \beta) = 9$

2. Let $f:\mathrm{R}\to\mathrm{R}$ be defined as $f(x)=\begin{cases}-\frac{4}{3}x^3+2x^2+3x, & x>0\\ 3xe^x, & x\le 0\end{cases}$. Then f is increasing function in the interval :

[JEE(Main)July2021]

माना : $f:\mathrm{R}\to\mathrm{R}, f(x)=\begin{cases}-\frac{4}{3}x^3+2x^2+3x, & x>0\\ 3xe^x, & x\le 0\end{cases}$ द्वारा परिभाषित है। तो निम्न में से किस अन्तराल में फलन f वर्धमान है?

[JEE(Main)July2021]

(A) $\left(-1,\frac{3}{2}\right)$ (B) $(0,2)$ (C) $(-3,-1)$ (D) $\left(-\frac{1}{2},2\right)$

Ans. A

Sol. $f'(x)=\begin{cases}-4x^2+4x+3 & ; x>0\\ 3e^x+3xe^x & ; x\le 0\end{cases}$

If function is increasing then

$f'(x)>0$

$-4x^2+4x+3>0$

$4x^2-4x-3\le 0$

$4x^2-6x+2x-3<0$

$2x(2x-3)+1(2x-3)\le 0$

$x\in\left(-\frac{1}{2},\frac{3}{2}\right)$ but here $x>0$

then $x\in\left(0,\frac{3}{2}\right)$ ....(1)

$3e^x+3xe^x>0$

$x+1>0$

$x>-1$ ....(2)

by (1) and (2)

$x\in\left(-1,\frac{3}{2}\right)$

3. Let $f(x) = 3\sin^4x + 10\sin^3x + 6\sin^2x - 3$, $x \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right]$. Then, f is : **[JEE(Main)July2021]**

(A) decreasing in $\left(0, \frac{\pi}{2}\right)$ (B) decreasing in $\left(-\frac{\pi}{6}, 0\right)$

(C) increasing in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (D) increasing in $\left(-\frac{\pi}{6}, \frac{\pi}{2}\right)$

माना : $f(x) = 3\sin^4x + 10\sin^3x + 6\sin^2x - 3$, $x \in \left[-\frac{\pi}{6}, \frac{\pi}{2}\right]$ है। तो f : **[JEE(Main)July2021]**

(A) $\left(0, \frac{\pi}{2}\right)$ में ह्रासमान है (B) $\left(-\frac{\pi}{6}, 0\right)$ में ह्रासमान है

(C) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में वर्धमान है (D) $\left(-\frac{\pi}{6}, \frac{\pi}{2}\right)$ में वर्धमान है

Ans. B

Sol. $f'(x) = 12\sin^3 + 30\sin^2 x + 12\sin x$

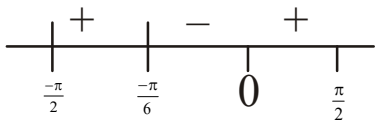
$= 6\sin x [2\sin^2 x + 5\sin x + 2\sin x]$

$= 6\sin x [(2\sin x + 1)(2\sin x + 2)]$

$= 6(\sin x + 2)(2\sin x + 1)(\sin x)$

$\sin x = 0, \Rightarrow x = 0$

$\sin x = \frac{-1}{2}, \quad \Rightarrow x = \frac{-\pi}{6}$



increasing in $x \in \left(-\frac{\pi}{2}, -\frac{\pi}{6}\right) \cup \left(0, \frac{\pi}{2}\right)$

decreasing in $x \in \left(-\frac{\pi}{6}, 0\right)$

4. Let $A = [a_{ij}]$ be a 3×3 matrix, where

$$a_{ij} = \begin{cases} 1 & , \text{ if } i = j \\ -x & , \text{ if } |i-j| = 1 \\ 2x+1, & \text{otherwise.} \end{cases}$$

Let a function $f: R \to R$ be defined as $f(x) = \det(A)$. Then the sum of maximum and minimum values of f on R is equal to : **[JEE(Main)July2021]**

माना $A = [a_{ij}]$ एक 3×3 का आव्यूह है, जहाँ

$$a_{ij} = \begin{cases} 1 & , \text{ यदि } i = j \\ -x & , \text{ यदि } |i-j| = 1 \\ 2x+1, & \text{ अन्यथा} \end{cases} \text{ है}$$

माना एक फलन $f: R \to R$, $f(x) = \det(A)$ द्वारा परिभाषित है। तो R पर f के अधिकतम तथा निम्नतम मानों का योगफल बराबर है : **[JEE(Main)July2021]**

(A) $\frac{88}{27}$ (B) $-\frac{88}{27}$ (C) $-\frac{20}{27}$ (D) $\frac{20}{27}$

Ans. B

Sol. $$A = \begin{bmatrix} 1 & -x & 2x+1 \\ -x & 1 & 2x+1 \\ 2x+1 & -x & 1 \end{bmatrix}$$

$\det(A) = 4x^3 - 4x^2 - 4x$

$f(x) = 4x^3 - 4x^2 - 4x$

$f'(x) = 12x^2 - 8x - 4$

$= 4(3x^2 - 2x - 1)$

$= 4(x-1)(3x+1)$

$x = 1, -1/3$ are extremum part

$f(1) = 4 - 4 - 4 = -4$

$$f\left(\frac{-1}{3}\right) = -\frac{-4}{27} - \frac{4}{9} + \frac{4}{3}$$

$$= 4\left(\frac{-1-3+9}{27}\right)$$

$$= \frac{20}{27}$$

$$f(1) + f(-1/3) = \frac{20}{27} - 4 = -\frac{88}{27}$$

5. Let 'a' be a real number such that the function $f(x) = ax^2 + 6x - 15$, $x \in R$, is increasing in $\left(-\infty, \frac{3}{4}\right)$ and decreasing in $\left(\frac{3}{4}, \infty\right)$. Then the function $g(x) = ax^2 - 6x + 15$, $x \in R$, has a : **[JEE(Main)July2021]**

(A) local minimum at $x = -\frac{3}{4}$ (B) local maximum at $x = \frac{3}{4}$

(C) local maximum at $x = -\frac{3}{4}$ (D) local minimum at $x = \frac{3}{4}$

माना 'a' एक वास्तविक संख्या है जिसके लिए फलन $f(x) = ax^2 + 6x - 15$, $x \in R$, अंतराल $\left(-\infty, \frac{3}{4}\right)$ में वर्धमान तथा अंतराल $\left(\frac{3}{4}, \infty\right)$ में ह्रासमान है। तो फलन $g(x) = ax^2 - 6x + 15$, $x \in R$ का : **[JEE(Main)July2021]**

(A) $x = -\frac{3}{4}$ पर स्थानीय निम्नतम है (B) $x = \frac{3}{4}$ पर स्थानीय अधिकतम है

(C) $x = -\frac{3}{4}$ पर स्थानीय अधिकतम है (D) $x = \frac{3}{4}$ पर स्थानीय निम्नतम है

Ans. C

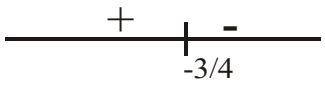
Sol. $f(x) = ax^2 + 6x + 15$

$f'(3/4) = 0$

$\Rightarrow a = -4$

$g(x) = -4x^2 - 6x + 15$

$g'(x) = -8x - 6$



$x = -3/4$ is a point of maxima

6. The sum of all the local minimum values of the twice differentiable function $f : R \to R$ defined by $f(x) = x^3 - 3x^2 - \frac{3f''(2)}{2}x + f''(1)$ is : **[JEE(Main)July2021]**

एक फलन $f : R \to R$, $f(x) = x^3 - 3x^2 - \frac{3f''(2)}{2}x + f''(1)$ द्वारा परिभाषित है तथा वह दो बार अवकलनीय है। इसके सभी स्थानीय न्यूनतम मानों का योग है : **[JEE(Main)July2021]**

(A) – 27 (B) 0 (C) 5 (D) – 22

Ans. A

Sol. $f(x) = x^3 - 3x^2 - \frac{3}{2}f''(2)x + f''(1)$

$f'(x) = 3x^2 - 6x - \frac{3}{2}f''(2)$

$f''(x) = 6x - 6$

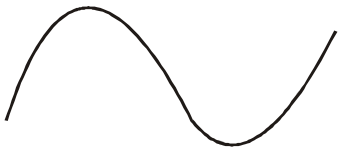
$f''(1) = 0, \quad f''(2) = 6$

$f(x) = x^3 - 3x^2 - 9x$

$f'(x) = 0$

$3x^2 - 6x - 9 = 0$

$x = -1$ and $x = 3$



Local minima at $x = 3$

$f(3) = 27 - 27 - 27 = -27$

7. The number of real roots of the equation $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ is :

[JEE(Main)July2021]

समीकरण, $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$ के वास्तविक मूलों की संख्या है : **[JEE(Main)July2021]**

(A) 4 (B) 6 (C) 1 (D) 2

Ans. D

Sol. $e^{6x} - e^{4x} - 2e^{3x} - 12e^{2x} + e^x + 1 = 0$

$e^{6x} - 2e^{3x} + 1 - e^{4x} - 12e^{2x} + e^x = 0$

$(e^{3x} - 1)^2 - e^x(e^{3x} - 1) - 12e^{2x} = 0$

$(e^{3x} - 1)^2 - e^x(e^{3x} - 1) - 12e^{2x} = 0$

let $e^{3x} - 1 = p \qquad e^x = q$

$p^2 - pq - 12q^2 = 0$

$p^2(p - 4q) + 39(p - 42) = 0$

$p(9 - 4q) + 3q(9 - 4q) = 0$

$(p + 3q)(p - 4q) = 0$

case-1

$p\ 3\ q = 0$

$e^{3x} - 1 + 3e^x = 0$

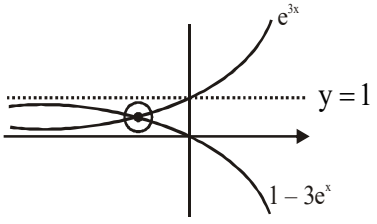
case-2

$p - 4q = 0$

$p = 4q$

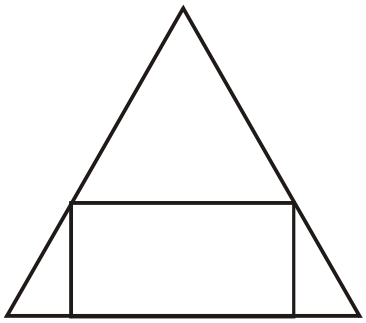
$e^{3x} - 1 = 4e^x$

$e^{3x} = 1 + 4e^x$



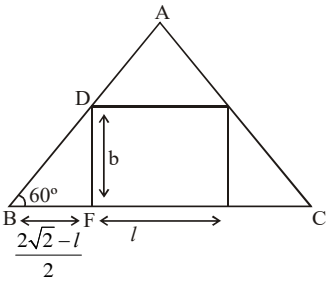
8. If a rectangle is inscribed in an equilateral triangle of side length $2\sqrt{2}$ as shown in the figure, then the square of the largest area of such a rectangle is _____. **[JEE(Main)July2021]**

यदि एक आयत $2\sqrt{2}$ भुजा की लंबाई वाले एक समबाहु त्रिभुज में अंकित है जैसा कि चित्र में दिखाया गया है, तो सबसे बड़े आयत का क्षेत्रफल वर्ग है _____ **[JEE(Main)July2021]**



Ans. 3

Ans.



In ΔDBF

$$\tan 60^\circ = \frac{2b}{2\sqrt{2}-l} \Rightarrow b = \frac{\sqrt{3}(2\sqrt{2}-l)}{2}$$

$$A = l \times b = l \times \frac{\sqrt{3}}{2}(2\sqrt{2}-l)$$

$$\frac{dA}{dl} = \frac{\sqrt{3}}{2}\left(2\sqrt{2}-2l\right) = 0$$

$$\sqrt{3}\left(\sqrt{2}-l\right) = 0$$

$$l = \sqrt{2}$$

$$A = l \times b = \sqrt{2}\left(\frac{\sqrt{3}}{2}\right)\left(2\sqrt{2}-\sqrt{2}\right)$$

$$= \frac{\sqrt{3}}{2}(2) = \sqrt{3}$$

$A^2 = 3$

9. An angle of intersection of the curves, $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and $x^2+y^2=ab$, $a>b$, is : **[JEE(Main)Aug.2021]**

वक्रों $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ तथा $x^2+y^2=ab$, $a>b$ का एक प्रतिच्छेदन कोण है : **[JEE(Main)Aug.2021]**

(A) $\tan^{-1}(2\sqrt{ab})$ (B) $\tan^{-1}\left(\frac{a+b}{\sqrt{ab}}\right)$ (C) $\tan^{-1}\left(\frac{a-b}{2\sqrt{ab}}\right)$ (D) $\tan^{-1}\left(\frac{a-b}{\sqrt{ab}}\right)$

Ans. D

Sol. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $x^2+y^2=ab$

$$\frac{2x}{a^2}+\frac{2yy'}{b^2}=0$$

$$\frac{yy'}{b^2}=-\frac{x}{a^2}$$

$$y'=-\frac{b^2}{a^2}\times\frac{x}{y}$$

$$y'_{(x_1,y_1)}=-\frac{b^2}{a^2}\left(\frac{x_1}{y_1}\right)$$

$$2x+2yy'=0$$

$$y'=-\frac{x}{y}=-\frac{x_1}{y_1}$$

$$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$$

$$\tan\theta=\left|\frac{\frac{x_1}{y_1}-\frac{b^2}{a^2}\left(\frac{x_1}{y_2}\right)}{1+\frac{b^2}{a^2}\left(\frac{x_1^2}{y_1^2}\right)}\right|$$

$$\tan\theta=\left|\frac{x_1y_1(a^2-b^2)}{a^2y_1^2+b^2x_1^2}\right|$$

$b^2x^2 + a^2y^2 = a^2b^2$

$b^2x_1^2 + a^2y_1^2 = a^2b^2$

$$\tan\theta = \left|\frac{x_1 y_1 (a^2 - b^2)}{a^2 b^2}\right|$$

$$y^2 = ab - x^2$$

$$\frac{x^2}{a^2} + \frac{ab - x^2}{b^2} = 1$$

$$\frac{x^2}{a^2} + \frac{a}{b} - \frac{x^2}{b^2} = 1$$

$$x^2\left(\frac{1}{a^2} - \frac{1}{b^2}\right) = 1 - \frac{a}{b}$$

$$x^2\left(\frac{b^2 - a^2}{a^2 b^2}\right) = \frac{b - a}{b}$$

$$x^2 = \frac{a^2 b}{b + a}$$

$$y^2 = ab - \frac{a^2 b}{a + b}$$

$$y^2 = \frac{a^2 b + ab^2 - a^2 b}{a + b} = \frac{ab^2}{a + b}$$

$$x^2 y^2 = \frac{a^3 b^3}{(a + b)^2}$$

$$xy = \frac{(a^3 b^3)^{1/2}}{a + b}$$

$$\tan\theta = \left|\frac{a^{3/2} b^{3/2}}{a + b} \times \frac{(a - b)(a + b)}{a^2 b^2}\right|$$

$$\tan\theta = \left|\frac{a - b}{\sqrt{ab}}\right|$$

$$\tan\theta = \frac{a - b}{\sqrt{ab}}$$

$$\theta = \tan^{-1}\left(\frac{a - b}{\sqrt{ab}}\right)$$

10. Let θ be the acute angle between the tangents to the ellipse $\frac{x^2}{9}+\frac{y^2}{1}=1$ and the circle $x^2+y^2=3$ at their point of intersection in the first quadrant. Then tanθ is equal to : **[JEE(Main)Aug.2021]**

माना दीर्घवृत्त $\frac{x^2}{9}+\frac{y^2}{1}=1$ तथा वृत्त $x^2+y^2=3$ के प्रथम चतुर्थांश में प्रतिच्छेदन बिन्दु पर स्पर्श रेखाओं के बीच न्यून कोण θ है, तब tanθ बराबर है : **[JEE(Main)Aug.2021]**

(A) 2 (B) $\frac{4}{\sqrt{3}}$ (C) $\frac{5}{2\sqrt{3}}$ (D) $\frac{2}{\sqrt{3}}$

Ans. D

Sol. $E:\frac{x^2}{9}+\frac{y^2}{1}=1$ ___(I)

$C:x^2+y^2=3$ ___(II)

For points of intersection,

$\frac{x^2}{9}+3-x^2=1$

$\Rightarrow 2=\frac{8x^2}{9}$

$x=\pm\frac{3}{2}$

$y=\pm\frac{\sqrt{3}}{2}$

point of intersection in the 1st quadrant

$=\left(\frac{3}{2},\frac{\sqrt{3}}{2}\right)$

<u>For slope of tangent to E :</u>

$\frac{2x}{9}+2y.\frac{dy}{dx}=0$

$\Rightarrow \frac{2}{9}\times\frac{3}{2}+2\times\frac{\sqrt{3}}{2}.m_1=0$

$\Rightarrow \frac{1}{3}=-\sqrt{3}\,m_1$

$m_1=\frac{-1}{3\sqrt{3}}$

For slope of tangent to C :

$$2x + 2y\frac{dy}{dx} = 0$$

$$\Rightarrow 2 \times \frac{3}{2} + 2 \times \frac{\sqrt{3}}{2} m_2 = 0$$

$$\Rightarrow 3 + \sqrt{3}\ m_2 = 0$$

$$\therefore m_2 = -\sqrt{3}$$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

$$= \left|\frac{\frac{-1}{3\sqrt{3}} + \sqrt{3}}{1 + \frac{1}{3}}\right|$$

$$= \left|\frac{8}{3\sqrt{3}} + \frac{3}{4}\right| = \frac{2}{\sqrt{3}}$$

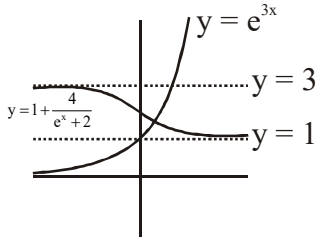
11. The number of real roots of the equation $e^{4x} + 2e^{3x} - e^x - 6 = 0$ is : **[JEE(Main)Aug.2021]**

समीकरण $e^{4x} + 2e^{3x} - e^x - 6 = 0$ के वास्तविक मूलों की संख्या है : **[JEE(Main)Aug.2021]**

(A) 0 (B) 2 (C) 4 (D) 1

Ans. D

Sol.



$$e^{3x}(e^x + 2) = e^x + 6$$

$$e^{3x} = 1 + \frac{4}{e^x + 2}$$

Number of real roots = 1

12. Let f be any continuous function on [0, 2] and twice differentiable on (0, 2). If f(0) = 0, f(1) = 1 and f(2) = 2, then: **[JEE(Main)Aug.2021]**

(A) f'(x) = 0 for some $x \in [0, 2]$ (B) f''(x) = 0 for some $x \in (0, 2)$

(C) f''(x) = 0 for all $x \in (0, 2)$ (D) f''(x) > 0 for some $x \in (0, 2)$

माना कोई फलन f अंतराल [0, 2] में संतत है तथा (0, 2) में दो बार अवकलनीय है। यदि f(0) = 0, f(1) = 1 तथा f(2) = 2 हैं तो :

[JEE(Main)Aug.2021]

(A) किसी $x \in [0, 2]$ के लिए $f'(x) = 0$ है

(B) सभी $x \in (0, 2)$ के लिए $f''(x) = 0$ है

(C) $x \in (0, 2)$ के लिए $f''(x) = 0$ है

(D) सभी $x \in (0, 2)$ के लिए $f''(x) > 0$ है

Ans. B

Sol. $f(0) = 0 \quad f(1) = 1 \quad f(2) = 2$

$g(x) = f(x) - x$

$g(0) = 0$, $g(1) = 0$, $g(2) = 0$ $\Rightarrow$ g(x) has three roots

$g'(x) = f'(x) - 1 \Rightarrow g'(x)$ has atleast two roots

$g''(x) = f''(x) \Rightarrow$ has atleast one roots

13. The function $f(x) = x^3 - 6x^2 + ax + b$ is such that $f(2) = f(4) = 0$. Consider two statements.

(S1) there exists $x_1, x_2 \in (2, 4)$, $x_1 < x_2$, such that $f'(x_1) = -1$ and $f'(x_2) = 0$.

(S2) there exists $x_3, x_4 \in (2, 4)$, $x_3 < x_4$, such that f is decreasing in $(2, x_4)$, increasing in $(x_4, 4)$ and $2f'(x_3) = \sqrt{3}\, f(x_4)$.

Then:

[JEE(Main)Aug.2021]

(A) both (S1) and (S2) are true

(B) (S1) is true and (S2) is false

(C) both (S1) and (S2) are false

(D) (S1) is false and (S2) is true

फलन $f(x) = x^3 - 6x^2 + ax + b$ ऐसा है कि $f(2) = f(4) = 0$ हैं। दो कथनों पर ध्यान दीजिए :

(S1) $x_1, x_2 \in (2, 4)$, $x_1 < x_2$ का अस्तित्व इस प्रकार है कि $f'(x_1) = -1$ तथा $f'(x_2) = 0$ हैं।

(S2) $x_3, x_4 \in (2, 4)$, $x_3 < x_4$ का अस्तित्व इस प्रकार है कि $(2, x_4)$ में f ह्रासमान है, $(x_4, 4)$ में f वर्धमान है तथा $2f'(x_3) = \sqrt{3}\, f(x_4)$ है। तब :

[JEE(Main)Aug.2021]

(A) (S1) तथा (S2) दोनों सत्य है

(B) (S1) सत्य है तथा (S2) असत्य है

(C) (S1) तथा (S2) दोनों असत्य है

(D) (S1) असत्य है तथा (S2) सत्य है

Ans. A

Sol. $f(x) = x^3 - 6x^2 + ax + b$

$f(2) = 0$

$2a + b = 16 \quad$ ____(I)

$f(4) = 0$

$4a + b = 32 \quad$ ____(II)

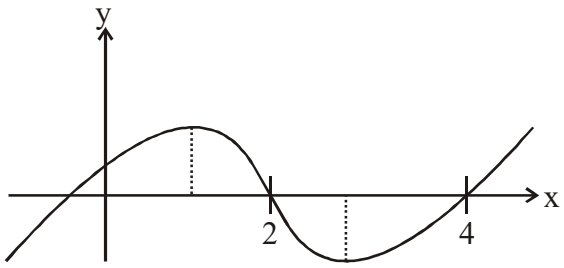
(II) – (I) $\quad a = 8$

$b = 0$

$\therefore f(x) = x^3 - 6x^2 + 8x = x(x-2)(x-4)$

$f'(x) = 3x^2 - 12 + 8$

$f'(x) = 0 \quad \therefore \quad x = 2 \pm \frac{2}{\sqrt{3}} = (\alpha, \beta)$



Slope at $f'(x_1) = 3x_1^2 - 12x_1 + 8 = -1$

$3[x_1^2 - 4x_1 + 3] = 0$

$x_1 = 1$ or $x_1 = 3$

$3 \in (2,4) \quad x_1 \in (2,4)$ s.t. $f'(x_1) = -1$

There exists $x_1 = 3$, $x_2 = 2 + \frac{2}{\sqrt{3}} \in (2, 4)$, $x_1 < x_2$, such that $f'(x_1) = -1$ and $f'(x_2) = 0$.

$\Rightarrow S_1$ is true.

$\because$ f(x) is decreasing in $(2, x_4)$ and increasing in $(x_4, 4)$

$f'(x_4) = 0$

$x_4 = 2 + \frac{2}{\sqrt{3}}$

$2f'(x_3) = \sqrt{3} f(x_4)$

$\Rightarrow 2[3x_3^2 - 12x_3 + 8] = \sqrt{3}\left(2 + \frac{2}{\sqrt{3}}\right)\left(\frac{2}{\sqrt{3}}\right)\left(\frac{2}{\sqrt{3}} - 2\right)$

$\Rightarrow 2(3x_3^2 - 12x_3 + 8) = 2\left(\frac{4}{3} - 4\right)$

$3x_3^2 - 12x_3 + 8 = -\frac{8}{3}$

$9x_3^2 - 36x_3 + 32 = 0$

$9x_3^2 - 24x_3 - 12x_3 + 32 = 0$

$(3x_3 - 4)(3x_3 - 8) = 0$

$x_3 = \frac{4}{3}$ or $x_3 = \frac{8}{3}$

since $2 < x_3 < x_3 \qquad \therefore x_3 = \frac{8}{3}$

There exists $x_3 = \frac{8}{3}$, $x_4 = 2 + \frac{2}{\sqrt{3}} \in (2, 4)$, $x_3 < x_4$, such that f is decreasing in $(2, x_4)$, increasing in $(x_4, 4)$ and

$2f'(x_3) = \sqrt{3}\, f(x_4)$.

S_2 is true.

14. The number of distinct real roots of the equation $3x^4 + 4x^3 - 12x^2 + 4 = 0$ is_____.

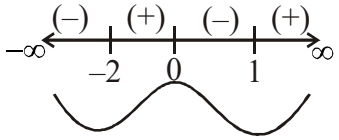
[JEE(Main)Aug.2021]

समीकरण $3x^4 + 4x^3 - 12x^2 + 4 = 0$ के भिन्न वास्तविक मूलों की संख्या है_____ **[JEE(Main)Aug.2021]**

Ans. 4

Sol. $f(x) = 3x^4 + 4x^3 - 12x^2 + 4$

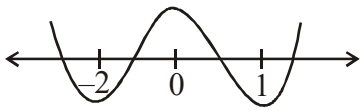
$f'(x) = 12x(x-1)(x+2)$

f'(x) = 

$f(-2) = -28(-)ve$

$f(0) = 4(+)ve$

$f(1) = -1(-)ve$

f(x) = 

No. of roots = 4

15. If 'R' is the least value of 'a' such that function $f(x) = x^2 + ax + 1$ is increasing on [1, 2] and 'S' is the greatest value of 'a' such that the function $f(x) = x^2 + ax + 1$ is decreasing on [1, 2], then the value of |R – S| is _____.

[JEE(Main)Aug.2021]

यदि 'a' का न्यूनतम मान, जिसके लिए फलन $f(x) = x^2 + ax + 1$, अंतराल [1, 2] पर वर्धमान हैं, 'R' है तथा 'a' का अधिकतम मान, जिसके लिए फलन $f(x) = x^2 + ax + 1$, अंतराल [1, 2] पर ह्रासमान है, तो |R – S| का मान _____

[JEE(Main)Aug.2021]

Ans. 2

Sol. $f(x) = x^2 + ax + 1$

$f'(x) = 2x + a$

For increasing $f'(x) \geq 0$

$2x + a \geq \theta$

$a \geq -2x$

$a \geq -2[1, 2]$

$a \geq [-4, -2]$

$R = -2$

For decreasing $f'(x) \leq 0$

$a \leq -2x$

$a \le [-4,-2]$

$a \le -4$

$S = -4$

$|R - S| = 2$

16. The local maximum value of the fucntion $f(x) = \left(\frac{2}{x}\right)^{x^2}$, $x > 0$, is : **[JEE(Main)Aug.2021]**

फलन $f(x) = \left(\frac{2}{x}\right)^{x^2}$, $x > 0$, का स्थानीय अधिकतम मान है : **[JEE(Main)Aug.2021]**

(A) $\left(\frac{4}{\sqrt{e}}\right)^{\frac{e}{4}}$ (B) $e^{\frac{2}{e}}$ (C) 1 (D) $\left(2\sqrt{e}\right)^{\frac{1}{e}}$

Ans. B

Sol. Let $y = \left(\frac{2}{x}\right)^{x^2} \quad (x > 0)$

$\ell n\, y = x^2 \ell n\left(\frac{2}{x}\right) = x^2\left(\ell n\, 2 - \ell n\, x\right)$

Differentiate both sides

$\frac{1}{y}.y' = 2x\left(\ell n 2 - \ell n x\right) + x^2\left(\frac{-1}{x}\right)$

$= x\left[\left(2\ell n 2 - 1\right) = 2\ell n x\right]$

$y'\left(\frac{2}{x}\right)^{x^2} x\left(\ell n \frac{4}{e} - 2\ell n x\right)$

$y' = 0$

$\Rightarrow \ell n\left(\frac{4}{e}\right) = 2\ell n x = \ell n x^2$

$\Rightarrow x^2 = \frac{4}{e}$

$\Rightarrow \frac{2}{x} = \sqrt{e}$

y is maximum at $x = \frac{2}{\sqrt{e}}$ as can be seen from sign change of y' across $x = \frac{2}{\sqrt{e}}$.

$$y_{max} = y\left(\frac{2}{\sqrt{e}}\right) = \left(\sqrt{e}\right)^{\frac{4}{e}} = e^{\frac{1}{2}\times\frac{4}{e}} = (e)^{\frac{2}{e}}$$

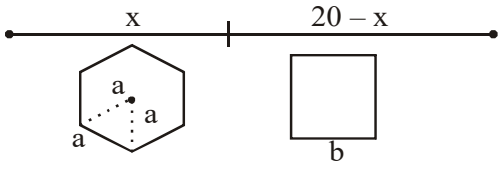
17. A wire of length 20 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a regular hexagon. Then the length of the side (in meters) of the hexagon, so that the combined area of the square and the hexagon is minimum, is : **[JEE(Main)Aug.2021]**

20 m लम्बाई की एक तार को दो भागों में काटा जाना है। एक भाग से एक वर्ग बनाना है तथा दूसरे भाग से एक सम षड्भुज बनाना है। तो दोनों वर्ग तथा षड्भुज के कुल क्षेत्रफल के न्यूनतम होने के लिए षड्भुज की भुजा की लम्बाई (मीटर में) है : **[JEE(Main)Aug.2021]**

(A) $\frac{10}{2+3\sqrt{3}}$ (B) $\frac{10}{3+2\sqrt{3}}$ (C) $\frac{5}{3+\sqrt{3}}$ (D) $\frac{5}{2+\sqrt{3}}$

Ans. B

Sol.



$6a = x$ $4b = 20 - x$

$a = x/6$ $b = 5 - x/4$

$$A = 6\cdot\left(\frac{\sqrt{3}}{4}a^2\right) + b^2 = \frac{3\sqrt{3}}{2}\cdot\frac{x^2}{36} + \left(5-\frac{x}{4}\right)^2$$

$$\frac{dA}{dx} = \frac{\sqrt{3}x}{12} + 2\left(5-\frac{x}{4}\right)\left(\frac{-1}{4}\right) = 0$$

$$\Rightarrow x = \frac{60}{3+2\sqrt{3}}$$

$$a = \frac{x}{6} = \frac{10}{3+2\sqrt{3}}$$

18. A box open from top is made from a rectangular sheet of dimension a×b by cutting squares each of side x from each of the four corners and folding up the flaps. If the volume of the box is maximum, then x is equal to: **[JEE(Main)Aug.2021]**

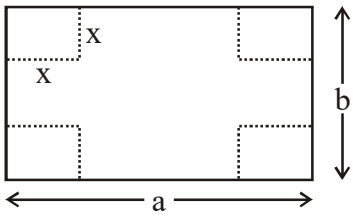
a×b (लम्बाई × चौड़ाई) की एक आयताकार चद्दर के प्रत्येक कोने से x भुजा के वर्ग काटकर तथा फलकों को मोड़कर ढ़क्कन रहित एक संदूक बनाया गया है। यदि संदूक का आयतन अधिकतम है, तो x बराबर है : **[JEE(Main)Aug.2021]**

(A) $\frac{a+b-\sqrt{a^2+b^2+ab}}{6}$ (B) $\frac{a+b-\sqrt{a^2+b^2-ab}}{6}$

(C) $\frac{a+b-\sqrt{a^2+b^2-ab}}{12}$ (D) $\frac{a+b+\sqrt{a^2+b^2-ab}}{6}$

Ans. B

Sol.



$v=(a-2x)(b-2x)x$

$v=(4x^2-2x(a+b)+ab)x$

$v=4x^3-2x^2(a+b)+abx$

$\frac{dv}{dx}=12x^2-4x(a+b)+ab=0$ $\begin{matrix}\alpha\\ \beta\end{matrix}$

$x=\frac{4(a+b)\pm\sqrt{16(a+b)^2-48ab}}{2\times12}$

$x=\frac{(a+b)\pm\sqrt{a^2+b^2-ab}}{6}$

Let $\alpha=\frac{(a+b)+\sqrt{a^2+b^2-ab}}{6}$

$\beta=\frac{(a+b)-\sqrt{a^2+b^2-ab}}{6}$



$\frac{dv}{dx}=12(x-\alpha)(x-\beta)$

here maximum value of at x = β

$x=\beta=\frac{(a+b)-\sqrt{a^2+b^2-ab}}{6}$

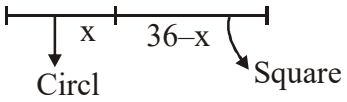
19. A wire of length 36 m is cut into two pieces, one of the pieces is bent to form a square and the other is bent to form a circle. If the sum of the areas of the two figures is minimum, and the circumference of the circle is k (meter), then $\left(\frac{4}{\pi}+1\right)k$ is equal to _____. **[JEE(Main)Aug.2021]**

36 m की एक तार दो भागों में काटा गया है। एक भाग को मोड़कर एक वर्ग बनाया गया है तथा दूसरे भाग को मोड़कर एक वृत्त बनाया गया है। यदि दोनों आकृतियों के क्षेत्रफल का योग निम्नतम है तथा वृत्त की परिधि k है, तो $\left(\frac{4}{\pi}+1\right)k$ बराबर है _____

[JEE(Main)Aug.2021]

Ans. 36

Sol.



$2\pi r = x \qquad 36 - x = 4a$

$r = \frac{x}{2\pi} \qquad a = 9 - \frac{x}{4}$

$S = \pi r^2 + a^2$

$S = \pi\left(\frac{x}{2\pi}\right)^2 + \left(9 - \frac{x}{4}\right)^2$

$S = \frac{x^2}{4\pi} + \left(9 - \frac{x}{4}\right)^2$

$\frac{ds}{dx} = \frac{x}{2\pi} - \frac{1}{2}\left(9 - \frac{x}{4}\right) = 0$

$\frac{x}{2\pi} + \frac{x}{8} = \frac{9}{2}$

$\frac{x}{8}\left(\frac{4}{\pi}+1\right) = \frac{9}{2}$

Ans. $= \left(\frac{4}{\pi}+1\right)k = \left(\frac{4}{\pi}+1\right)x$

$= \left(\frac{9}{2}\right)8 = 36$

20. Let f(x) be a cubie polynomical with f(1) = –10, f(–1) = 6, and has a local minima at x = 1, and f'(x) has a local minima at x = –1. Then f(3) is equal to_____. **[JEE(Main)Aug.2021]**

माना f(x) एक त्रिघातीय बहुपद है जिसके लिए f(1) = –10, f(–1) = 6 हैं, तथा f का एक स्थानीय निम्नतम बिन्दु x = 1 है और f'(x) का एक स्थानीय निम्नतम बिन्दु x = –1 है तो f(3) बराबर है _____ **[JEE(Main)Aug.2021]**

Ans. 22

Sol. f(1) = –10 f(–1) = 6

f''(x) = a(x + 1)

$$f'(x) = \frac{ax^2}{2} + ax + c$$

$$f'(1) = 0 \quad \Rightarrow \quad c = \frac{-3a}{2}$$

$$f'(x) = \frac{ax^2}{2} + ax - \frac{3a}{2}$$

$$f(x) = \frac{ax^3}{6} + \frac{ax^2}{2} - \frac{3ax}{2} + c$$

$f(1) = -10$

$f(-1) = 6$

$= 1\ c = -5\ \ a = 6$

$f(x) = x^3 + 3x^2 - 9x - 5$

$f(3) = 22$

21. The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is:
गोले के आकार के एक गुब्बारे में हवा भरने के दौरान उसके पृष्ठ का क्षेत्रफल एक स्थिर दर से बढ़ता है। यदि आरंभ में गुब्बारे की त्रिज्या 3 इकाई है तथा 5 सैकण्ड पश्चात् यह 7 इकाई हो जाती है, तो 9 सैकण्ड पश्चात् इसकी त्रिज्या होगी :

[JEE(Main)June2022]

(A) 9 unit (B) 10 unit (C) 11 unit (D) 12 unit

Ans. A

Sol. $S = 4\pi r^2$

$$\frac{ds}{dt} = k$$

$$8\pi r\frac{dr}{dt} = k$$

$$\int r dr = \int \frac{k}{8\pi} dt$$

$$\frac{r^2}{2} = \frac{kt}{8\pi} + c$$

$$\text{at } t = 0,\ r = 3 \Rightarrow c = \frac{9}{2}$$

$$\Rightarrow r^2 = \frac{kt}{4\pi} + 9$$

$$\text{at } t = 5,\ r = 7 \Rightarrow k = 32\pi$$

$$\Rightarrow r^2 = 8t + 9$$

at $t = 9 \Rightarrow r^2 = 81$

$\Rightarrow r = 9$

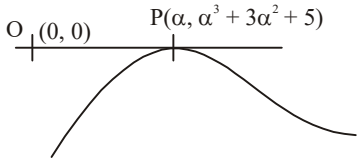
22. If the tangent at the point(x_1, y_1) on the curve $y = x^3 + 3x^2 + 5$ passes through the origin, then (x_1, y_1) does NOT lie on the curve : **[JEE(Main)June2022]**

यदि वक्र $y = x^3 + 3x^2 + 5$ के बिन्दु(x_1, y_1) पर स्पर्ष रेखा मूल बिन्दु से होकर जाती है, तो (x_1, y_1) निम्न में से किस वक्र पर स्थित नहीं है : **[JEE(Main)June2022]**

(A) $x^2 + \frac{y^2}{81} = 2$ (B) $\frac{y^2}{9} - x^2 = 8$ (C) $y = 4x^2 + 5$ (D) $\frac{x}{3} - y^2 = 2$

Ans. D

Sol.



equate slope of OP

$$3\alpha^2 + 6\alpha = \frac{\alpha^3 + 3\alpha^2 + 5}{\alpha}$$

$2\alpha^3 + 3\alpha^2 = 5 \Rightarrow \alpha = 1$

P(1, 9)

P does not lie on $\frac{x}{3} - y^2 = 2$

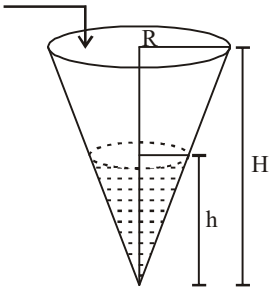
23. Water is being filled at the rate of 1 cm^3/sec in a right circular conical vessel (vertex downwards) of height 35 cm and diameter 14 cm. When the height of the water level is 10cm, the rate (in cm^2/sec) at which the wet conical surface area of the vessel increases is : **[JEE(Main)June2022]**

35 सेमी ऊँचाई तथा 14 सेमी व्यास के एक लंब वृत्तीय शंक्वाकार बर्तन, जिसका शीर्ष नीचे की ओर है, में 1 cm^3/sec की दर से पानी भरा जा रहा है। जब पानी की सतह की ऊँचाई 10 सेमी है, तब बर्तन के भीगे हुए शंक्वाकार पृष्ठीय क्षेत्रफल की वृद्धि दर (सेमी^{2}/सै. में) है : **[JEE(Main)June2022]**

(A) 5 (B) $\frac{\sqrt{21}}{5}$ (C) $\frac{\sqrt{26}}{5}$ (D) $\frac{\sqrt{26}}{10}$

Ans. C

Sol.



$$\frac{R}{r}=\frac{H}{h}\Rightarrow\frac{7}{r}=\frac{35}{h}$$

$$\Rightarrow h=5r$$

$$V=\frac{1}{3}\pi r^2h=\frac{5\pi}{3}r^3$$

$$\frac{dV}{dt}=5\pi r^2\frac{dr}{dt}=1$$

$$\frac{dr}{dt}=\frac{1}{5\pi r^2}$$

$$S=\pi r\ell=\pi r\sqrt{r^2+h^2}=\pi r.\sqrt{r^2+25r^2}$$

$$S=\sqrt{26}\pi r^2$$

$$\frac{ds}{dt}=\sqrt{26}\pi.2r\frac{dr}{dt}$$

$$\frac{ds}{dt}=2\sqrt{26}\,\pi.r.\frac{1}{5\pi r^2}$$

$$\frac{ds}{dt}=\frac{2\sqrt{26}}{5r}\qquad\text{(when } h=10\text{cm, } r=2\text{cm)}$$

$$\frac{ds}{dt}=\frac{\sqrt{26}}{5}$$

24. If the angle made by the tangent at the point (x_0, y_0) on the curve $x = 12\,(t + \sin t \cos t)$, $y = 12(1 + \sin t)^2$, $0 < t < \frac{\pi}{2}$, with the positive x-axis is $\frac{\pi}{3}$, then y_0 is equal to : **[JEE(Main)June2022]**

यदि वक्र $x = 12\,(t + \sin t \cos t)$, $y = 12(1 + \sin t)^2$, $0 < t < \frac{\pi}{2}$ के बिन्दु (x_0, y_0) पर स्पर्ष रेखा धनात्मक x-अक्ष से $\frac{\pi}{3}$ का कोण बनाती है, तो y_0 बराबर है : **[JEE(Main)June2022]**

(A) $6\left(3+2\sqrt{2}\right)$ (B) $3\left(7+4\sqrt{3}\right)$ (C) 27 (D) 48

Ans. C

Sol. $$\frac{dy}{dx}=\frac{\left(\frac{dy}{dt}\right)}{\left(\frac{dx}{dt}\right)}=\frac{12.2(1+\sin t).\cos t}{12(1+\sin t.(-\sin t)+\cos t.\cos t)}$$

$$=\frac{2(1+\sin t)\cos t}{2\cos^2 t}=\tan\left(\frac{\pi}{3}\right)=\sqrt{3}$$

$$\Rightarrow 1+\sin t=\sqrt{3}\ \cos t$$

$$\frac{\sqrt{3}}{2}\cos t-\frac{1}{2}.\sin t=\frac{1}{2}$$

$$\cos\left(t+\frac{\pi}{6}\right)=\cos\frac{\pi}{3}$$

$$t=\frac{\pi}{6}$$

$$y_0=12\left(1+\sin\frac{\pi}{6}\right)^2=27$$

25. Let S be the set of all the natural numbers, for which the line $\frac{x}{a}+\frac{y}{b}=2$ is a tangent to the curve $\left(\frac{x}{a}\right)^n+\left(\frac{y}{b}\right)^n=2$ at the point (a, b), ab ≠ 0. Then : **[JEE(Main)June2022]**

माना सभी धन पूर्णांकों, जिनके लिए रेखा $\frac{x}{a}+\frac{y}{b}=2$, वक्र $\left(\frac{x}{a}\right)^n+\left(\frac{y}{b}\right)^n=2$ के बिंदु (a, b), ab ≠ 0 पर स्पर्ष रेखा है, का समुच्चय S है। तो : **[JEE(Main)June2022]**

(A) $S=\phi$ (B) $n(S)=1$ (C) $S=\{2k : k\in N\}$ (D) $S=N$

Ans. D

Sol. $$\left(\frac{x}{a}\right)^n+\left(\frac{y}{b}\right)^n=2$$

Slope of tangent at (a, b)

$$n.\left(\frac{x}{a}\right)^{n-1}.\frac{1}{a}+n\left(\frac{x}{b}\right)^{n-1}.\frac{1}{b}\frac{dy}{dx}=0$$

$$\left.\frac{dy}{dx}\right|_{(a,b)}=-\frac{b}{a}$$

∴ Equation of tangent

$$y-b=-\frac{b}{a}(x-a)$$

$$\frac{x}{a}+\frac{y}{b}=2\forall n\in N$$

26. Consider a cuboid of sides 2x, 4x and 5x and a closed hemisphere of radius r. If the sum of their surface areas is a constant k, then the ratio x : r, for which the sum of their volumes is maximum, is :

[JEE(Main)June2022]

भुजाओं 2x, 4x तथा 5x के एक घनाभ तथा त्रिज्या r के एक बंद गोलार्ध का विचार कीजिए। यदि उनके पृष्ठीय क्षेत्रफलों का योग एक अचर k है, तो उनके आयतन का योग अधिकतम होने के लिए अनुपात x : r है : **[JEE(Main)June2022]**

(A) 2 : 5 (B) 19 : 45 (C) 3 : 8 (D) 19 : 15

Ans. B

Sol. T.S.A of cuboid = $2\left(8x^2+20x^2+10x^2\right)=76x^2$

T.S.A of closed Hemis pheve = $3\pi r^2$

$$76x^2+3\pi r^2=k$$

$$152x\frac{dx}{dr}+6\pi r=0$$

$$\frac{dx}{dr}=-\frac{6\pi r}{152x}$$

Volume = $40x^3+\frac{2}{3}\pi r^3$

$$\frac{dv}{dr}=120x^2\frac{dx}{dr}+2\pi r^2=0$$

$$120x^2\times\left(\frac{-6\pi r}{152x}\right)+2\pi r^2=0$$

$$120x\times\left(+\frac{6}{152}\right)=+2r$$

$$\frac{x}{r}=\frac{2\times152}{6\times120}$$

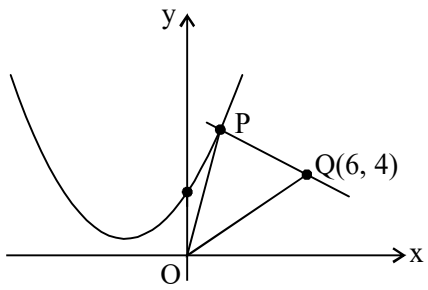
$$\frac{x}{r}=\frac{19}{3\times15}=\frac{19}{45}$$

27. Let *l* be a line which is normal to the curve $y=2x^2+x+2$ at a point P on the curve. If the point Q(6, 4) lies on the line *l* and O is origin, then the area of the triangle OPQ is equal to _____. **[JEE(Main)June2022]**

माना l एक रेखा है जो वक्र $y = 2x^2 + x + 2$ के बिन्दु P पर अभिलंब है। यदि बिन्दु Q(6, 4), रेखा l पर है तथा O मूल बिन्दु है, तो त्रिभुज OPQ का क्षेत्रफल बराबर है _____. **[JEE(Main)June2022]**

Ans. Official Answer NTA (13)

Sol. $y = 2x^2 + x + 2$



$$\frac{dy}{dx} = 4x + 1$$

Let P be (h, k), then normal at P is

$$y - k = -\frac{1}{4h+1}(x - h)$$

This passes through Q (6,4)

$$\therefore 4 - k = -\frac{1}{4h+1}(6 - h)$$

$$\Rightarrow (4h+1)(4-k) + 6 - h = 0$$

Also $k = 2h^2 + h + 2$

$\therefore (4h + 1)(4 - 2h^2 - h - 2) + 6 + h = 0$

$\Rightarrow 4h^3 - 3h^2 + 3h - 8 = 0$

$\Rightarrow h = 1, k = 5$

Now area of ΔOPQ will be $= \frac{1}{2}\begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 5 \\ 1 & 6 & 4 \end{vmatrix} = 13$

28. For the function $f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5$, $x > 1$, which one of the following is NOT correct?

[JEE(Main)June2022]

(A) f is increasing in (1, 2) and decreasing in $(2, \infty)$

(B) $f(x) = -1$ has exactly two solutions

(C) $f'(e) - f''(2) < 0$

(D) $f(x) = 0$ has a root in the interval $(e, e + 1)$

फलन $f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5$, $x > 1$, के लिए निम्न में से कौनसा एक सही नहीं है?

[JEE(Main)June2022]

(A) f अन्तराल (1, 2) में वर्धमान तथा $(2, \infty)$ में ह्रासमान है

(B) $f(x) = -1$ के ठीक दो हल हैं

(C) $f'(e) - f''(2) < 0$ है

(D) $f(x) = 0$ का एक मूल अंतराल $(e, e+1)$ में है

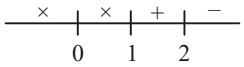
Ans. C

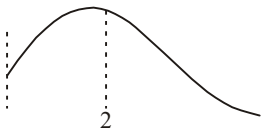
Sol. $f(x) = 4\log_e(x-1) - 2x^2 + 4x + 5$

$$f'(x) = \frac{4}{x-1} - 4x + 4$$

$$= -\frac{4(x)(x-2)}{(x-1)}$$

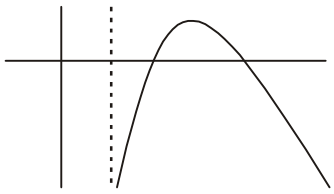
$$f''(x) = \frac{-4}{(x-1)^2} - 4$$





$\Rightarrow$ Option A is true

graph of f(x)



$\Rightarrow$ Option B is also true

$$f'(e) = \frac{4}{e-1} - 4(e-1)$$

$$f''(2) = -8$$

$$f'(e) - f''(2) > 0$$

$\Rightarrow$ Option C is false

29. The number of distinct real roots of the equation $x^7 - 7x - 2 = 0$ is : **[JEE(Main)June2022]**

समीकरण $x^7 - 7x - 2 = 0$ के भिन्न वास्तविक मूलों की संख्या है : **[JEE(Main)June2022]**

(A) 5 (B) 7 (C) 1 (D) 3

Ans. D

Sol. $x^7 - 7x - 2 = 0$

Consider

$f(x) = x^7 - 7x - 2$

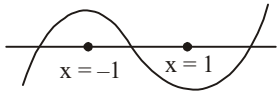
$f'(x) = 7x^6 - 7$

$f'(x) = 7(x^6 - 1)$

$= 7(x^2 - 1)(x^4 + x^2 + 1)$

$= 7(x+1)(x-1)(x^4 + x^2 + 1)$

If we scatch graph of f(x)



Clearly number of solutions are 3

30. Let λ^* be the largest value of λ for which the function $f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$ is increasing for all $x \in R$. Then $f_{\lambda^*}(1) + f_{\lambda^*}(-1)$ is equal to : **[JEE(Main)June2022]**

माना λ^* का अधिकतम मान, जिसके लिए फलन $f_\lambda(x) = 4\lambda x^3 - 36\lambda x^2 + 36x + 48$ सभी $x \in R$ के लिए वर्धमान है, λ^* है। तो $f_{\lambda^*}(1) + f_{\lambda^*}(-1)$ बराबर है : **[JEE(Main)June2022]**

(A) 36 (B) 48 (C) 64 (D) 72

Ans. D

Sol. $f_\lambda(x) = 4\lambda x^3 - 36dx^2 + 36x + 48$

$f_\lambda'(x) = 12\lambda x^2 - 72\lambda x + 36$

$f_\lambda'(x) = 12(\lambda x^2 - 6\lambda x + 3)$

since function is increasing

$\lambda > 0$ ——(1)

$\lambda < 0$

$36\lambda^2 - 12\lambda \le 0$ ——(2)

by (1) and (2)

$$\lambda \in \left(0, \frac{1}{3}\right]$$

$$f_{\frac{1}{3}}(x) = \frac{4}{3}x^3 - 12x^2 + 36x + 48$$

$$f_{\frac{1}{3}}(1)=\frac{4}{3}-12+36+48$$

$$f_{\frac{1}{3}}(-1)=\frac{-4}{3}-12-36+48$$

$$f_{\frac{1}{3}}(1)+f_{\frac{1}{3}}(-1)=72$$

31. Let $f: R \to R$ and $g: R \to R$ be two functions defined by $f(x)=\log_e(x^2+1)-e^{-x}+1$ and $g(x)=\frac{1-2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right)>f\left(g\left(\alpha-\frac{5}{3}\right)\right)$ holds?

[JEE(Main)June2022]

माना दो फलन $f: R \to R$ तथा $g: R \to R$, $f(x)=\log_e(x^2+1)-e^{-x}+1$ तथा $g(x)=\frac{1-2e^{2x}}{e^x}$ द्वारा परिभाषित हैं। तो α के निम्न में से किस परिसर (range) के लिए असमिका $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right)>f\left(g\left(\alpha-\frac{5}{3}\right)\right)$ संतुष्ट होती है?

[JEE(Main)June2022]

(A) (2, 3) (B) (–2, –1) (C) (1, 2) (D) (–1, 1)

Ans. A

Sol. $f(x)=\log_e(x^2+1)-e^{-x}+1$

$\Rightarrow f'(x)=\frac{2x}{x^2+1}+e^{-x}>0 \ \forall \ x \in R$

$\Rightarrow$ f is stricitly increasing

$g(x)=\frac{1-2e^{2x}}{e^x}=e^{-x}-2e^x$

$\Rightarrow g'(x)=-\left(2e^x+e^{-x}\right)<0 \ \forall \ x \in R$

$\Rightarrow$ g is decreasing

Now $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right)>f\left(g\left(\alpha-\frac{5}{3}\right)\right)$

$\Rightarrow g\left(\frac{(\alpha-1)^2}{3}\right)>g\left(\alpha-\frac{5}{3}\right)$

$$\Rightarrow \frac{(\alpha-1)^2}{3} < \alpha - \frac{5}{3}$$

$$\Rightarrow \alpha^2 - 5\alpha + 6 < 0$$

$$\Rightarrow (\alpha-2)(\alpha-3) < 0$$

$$\Rightarrow \alpha \in (2,3)$$

32. Let $f(x) = 2\cos^{-1}x + 4\cot^{-1}x - 3x^2 - 2x + 10$, $x \in [-1, 1]$. If $[a, b]$ is the range of the function f, then $4a - b$ is equal to : **[JEE(Main)June2022]**

माना $f(x) = 2\cos^{-1}x + 4\cot^{-1}x - 3x^2 - 2x + 10$, $x \in [-1, 1]$ है। यदि फलन f का परिसर $[a, b]$ है, तो $4a - b$ बराबर है: **[JEE(Main)June2022]**

(A) 11 (B) $11 - \pi$ (C) $11 + \pi$ (D) $15 - \pi$

Ans. B

Sol. $$f'(X) = \frac{-2}{\sqrt{1-x^2}} - \frac{4}{1+x^2} - 6x - 2$$

$$= -2\left[\frac{1}{\sqrt{1-x^2}} + \frac{2}{1+x^2} + 3x + 1\right]$$

$f'(x) < 0 \Rightarrow f(x)$ is a dec. function

$f(1) = \pi + 5$

$f(-1) = 5\pi + 5$

Range : $[a, b] \equiv [\pi + 5, 5\pi + 5]$

$a = \pi + 5$, $b = 5\pi + 5 \Rightarrow 4a - b = 11 - \pi$.

33. The number of distinct real roots of $x^4 - 4x + 1 = 0$ is : **[JEE(Main)June2022]**

समीकरण $x^4 - 4x + 1 = 0$ के भिन्न वास्तविक मूलों की संख्या है : **[JEE(Main)June2022]**

(A) 4 (B) 2 (C) 1 (D) 0

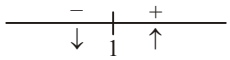
Ans. B

Sol. $x^4 - 4x + 1 = 0$

$f(x) = x^4 - 4x + 1$

$$f'(x) = 4x^3 - 4$$

$$f'(x) = 4(x^3 - 1) = 4(x-1)(x^2 + x + 1)$$



So f(x) will have a minima at x = 1

f(x) = 1 – 4 + 1 = –2

so there will be 2 roots

34. The number of real solutions of $x^7 + 5x^3 + 3x + 1 = 0$ is equal to _____. **[JEE(Main)June2022]**

$x^7 + 5x^3 + 3x + 1 = 0$ के वास्तविक हलों की संख्या है _____ **[JEE(Main)June2022]**

(A) 0 (B) 1 (C) 3 (D) 5

Ans. B

Sol. Let $f(x) = x^7 + 5x^3 + 3x + 1$

$f'(x) = 7x^6 + 15x^2 + 3 > 0$

So are real root as it is odd degree polynomial

35. Let f and g be twice differentiable even functions on (–2, 2) such that

$$f\left(\frac{1}{4}\right)=0, f\left(\frac{1}{2}\right)=0, f(1)=1 \text{ and } g\left(\frac{3}{4}\right)=0, g(1)=2$$

Then, the minimum number of solutions of $f(x)g''(x)+f'(x)g'(x)=0$ in (–2, 2) is equal to _____.

[JEE(Main)June2022]

माना (–2, 2) पर दो बार अवकलनीय सम फलनों f तथा g के लिए

$$f\left(\frac{1}{4}\right)=0, f\left(\frac{1}{2}\right)=0, f(1)=1 \text{ तथा } g\left(\frac{3}{4}\right)=0, g(1)=2$$

तो (–2,2) में $f(x)g''(x)+f'(x)g'(x)=0$ के हलों की न्यूनतम संख्या है _____. **[JEE(Main)June2022]**

Ans. 4

Sol. Let $h(x) = f(x)\, g'(x) \rightarrow 5$ roots

$\because$ f(x) is evne $\Rightarrow f\left(\frac{1}{4}\right)=f\left(\frac{1}{2}\right)=f\left(-\frac{1}{2}\right)=f\left(\frac{1}{4}\right)=0$

g(x) is even $\Rightarrow g\left(\frac{3}{4}\right)=g\left(-\frac{3}{4}\right)=0$

g'(x) = 0 has minimum one root

h'(x) has at last 4 roots

36. The sum of absolute maximum and absolute minimum values of the function $f(x) = |2x^2 + 3x – 2| + \sin x \cos x$ in the interval [0, 1] is : **[JEE(Main)June2022]**

अंतराल [0, 1] में फलन $f(x) = |2x^2 + 3x - 2| + \sin x \cos x$ के निरपेक्ष उच्चतम तथा निरपेक्ष निम्नतम मानों का योगफल है :

[JEE(Main)June2022]

(A) $3+\frac{\sin(1)\cos^2\left(\frac{1}{2}\right)}{2}$

(B) $3+\frac{1}{2}(1+2\cos(1))\sin(1)$

(C) $5+\frac{1}{2}(\sin(1)+\sin(2))$

(D) $2+\sin\left(\frac{1}{2}\right)\cos\left(\frac{1}{2}\right)$

Ans. B

Sol. $f(x)\begin{cases}\rightarrow 2x^2+3x-2+\frac{\sin 2x}{2} & \frac{1}{2}\le x\le 1\\ \rightarrow -(2x^2+3x-2)+\frac{\sin 2x}{2} & 0\le x<\frac{1}{2}\end{cases}$

$f'(x)\begin{cases}\rightarrow 4x+3+\cos(2x) & \frac{1}{2}<x\le 1\\ \rightarrow -4x-3+\cos 2x & 0\le x<\frac{1}{2}\end{cases}$

$f'(x)<0 \quad \forall x\in\left[0,\frac{1}{2}\right)$

$f'(x)>0 \quad \forall\, x\in\left(\frac{1}{2},1\right]$

$f(x)_{min}=f\left(\frac{1}{2}\right)=\frac{\sin(1)}{2}$

$f(x)_{max}=\max\{f(0),f(1)\}$

$=f(1)=3+\sin 1\cos 1$

$\text{sum}=3+\sin 1\cos 1+\frac{\sin 1}{2}$

37. Let the maximum area of the triangle that can be inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{4}=1$, $a>2$, having one of its vertices at one end of the major axis of the ellipse and one of its sides parallel to the y-axis, be $6\sqrt{3}$. Then the eccentricity of the ellipse is :

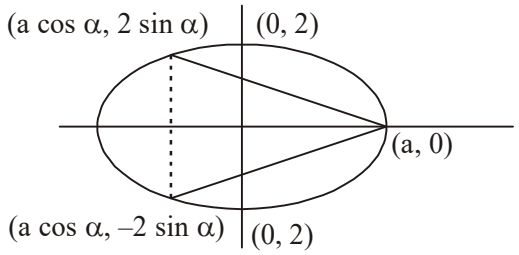
[JEE(Main)June2022]

माना दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{4}=1$, $a>2$ के अन्तर्गत त्रिभुज, जिसका एक शीर्ष दीर्घवृत्त के दीर्घअक्ष के एक सिरे पर है तथा जिसकी एक भुजा y-अक्ष के समांतर है, का अधिकतम क्षेत्रफल $6\sqrt{3}$ है। तो दीर्घवृत्त की उत्केन्द्रता है : **[JEE(Main)June2022]**

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{\sqrt{3}}{4}$

Ans. A

Sol.



$$\text{Are of triangle} = \frac{1}{2}\left|4\sin\alpha\left(a-a\cos\alpha\right)\right|$$

$$=\frac{1}{2}\left|4a\sin\alpha\left(1-\cos\alpha\right)\right|$$

Let consider $f(\alpha)=\sin\alpha(1-\cos\alpha)$

Minimum value of $f(\alpha)$ will be at $\alpha=\frac{2\pi}{3}$

$$f(\alpha)_{max}=\frac{3\sqrt{3}}{4}$$

$$\text{maximum area of } \Delta=\frac{1}{2}\left|4a\frac{3\sqrt{3}}{4}\right|=6\sqrt{3}$$

$$e=\sqrt{1-\frac{b^2}{a^2}} \qquad a=4$$

$$e=\sqrt{1-\frac{4}{16}}$$

$$=\sqrt{1-\frac{1}{4}}=\frac{\sqrt{3}}{2}$$

38. The sum of the absolute minimum and the absolute maximum values of the function $f(x)=|3x-x^2+2|-x$ in the interval $[-1,2]$ is : **[JEE(Main)June2022]**

फलन $f(x)=|3x-x^2+2|-x$ के अंतराल $[-1,2]$ में निरपेक्ष न्यूनतम तथा निरपेक्ष उच्चतम मानों का योग है : **[JEE(Main)June2022]**

(A) $\frac{\sqrt{17}+3}{2}$ (B) $\frac{\sqrt{17}+5}{2}$ (C) 5 (D) $\frac{9-\sqrt{17}}{2}$

Ans. A

Sol.

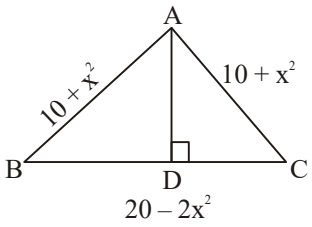
39. The lengths of the sides of a triangle are $10 + x^2$, $10 + x^2$ and $20 - 2x^2$. If for $x = k$, the area of the triangle is maximum, then $3k^2$ is equal to : **[JEE(Main)June2022]**

एक त्रिभुज की भुजाओं की लंबाई $10 + x^2$, $10 + x^2$ तथा $20 - 2x^2$ है। यदि $x = k$ के लिए त्रिभुज का क्षेत्रफल अधिकतम है, तो $3k^2$ बराबर है : **[JEE(Main)June2022]**

(A) 5 (B) 8 (C) 10 (D) 12

Ans. C

Sol. $BD = 10 - x^2$



$$AD = \sqrt{(10+x^2)-(10-x^2)^2}$$

$$AD = \sqrt{(10+10)(x^2+x^2)}$$

$$AD = \sqrt{40}x$$

$$\text{Area} = \frac{1}{2}AD \times BC = \frac{1}{2}\sqrt{40}x\left(20-2x^2\right)$$

$$A = \sqrt{40}\left(10x - x^3\right)$$

$$\frac{dA}{dx} = 0$$

$$\sqrt{40}\left(10-3x^2\right) = 0$$

$$x^2 = \frac{10}{3}$$

$3x^2 = 10$

$3k^2 = 10$

40. A wire of length 22 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into an equilateral triangle. Then, the length of the side of the equilateral triangle, so that the combined area of the square and the equilateral triangle is minimum, is : **[JEE(Main)June2022]**

एक 22 मी. लंबे तार को दो टुकड़ों में विभक्त किया जाना है। एक टुकड़े से एक वर्ग तथा दूसरे से एक समबाहु त्रिभुज बनाया जाना है। समबाहु त्रिभुज की भुजा की लंबाई, जिसके लिए वर्ग तथा समबाहु त्रिभुज का सम्मिलित क्षेत्रफल न्यूनतम हो, है :

[JEE(Main)June2022]

(A) $\frac{22}{9+4\sqrt{3}}$ (B) $\frac{66}{9+4\sqrt{3}}$ (C) $\frac{22}{4+9\sqrt{3}}$ (D) $\frac{66}{4+9\sqrt{3}}$

Ans. B

Sol. Let x be the length out from a wire of length 22 m.

and made of wire an equilateral Δ whose side is b.

$3b = x$ and remaining length $= 22 - x$

$$b = \frac{x}{3}$$

Let a be the side of square

$$4a = 22 - x \Rightarrow a = \frac{22-x}{4}$$

A = combined area of the square and equilateral Δ.

$$= \frac{\sqrt{3}}{4}b^2 + a^2$$

$$= \frac{\sqrt{3}}{4}\frac{x^2}{9} + \frac{(22-x)^2}{16} = \frac{\sqrt{3}}{36}x^2 + \frac{1}{16}x^2 - \frac{44x}{16} + \frac{(22)^2}{16}$$

$$A = \left(\frac{\sqrt{3}}{36} + \frac{1}{16}\right)x^2 - \frac{11}{4}x + \frac{484}{16}$$

$$A_{min} \text{ at } x = \frac{\frac{11}{4}}{2\left(\frac{\sqrt{3}}{36} + \frac{1}{16}\right)}$$

$$b = \frac{x}{3} = \frac{1}{3}.\frac{11.4.4.9}{4.2\left(4\sqrt{3}+9\right)}$$

$$b = \frac{66}{9+4\sqrt{3}}$$

41. Let $f: R \to R$ be a function defined by $f(x) = (x-3)^{n_1}(x-5)^{n_2}, n_1, n_2 \in N$. Then, which of the following is NOT true? **[JEE(Main)June2022]**

(A) For $n_1 = 3, n_2 = 4$, there exists $\alpha \in (3, 5)$ where f attains local maxima

(B) For $n_1 = 4, n_2 = 3$, there exists $\alpha \in (3, 5)$ where f attains local minima

(C) For $n_1 = 3, n_2 = 5$, there exists $\alpha \in (3, 5)$ where f attains local maxima

(D) For $n_1 = 4, n_2 = 6$, there exists $\alpha \in (3, 5)$ where f attains local maxima

माना फलन $f : R \to R, f(x) = (x-3)^{n_1}(x-5)^{n_2}, n_1, n_2 \in N$ द्वारा परिभाषित है। तो निम्न में से कौनसा सत्य नहीं है?

[JEE(Main)June2022]

(A) $n_1 = 3, n_2 = 4$, के लिए ऐसे $\alpha \in (3, 5)$ का अस्तित्व है, जहाँ f स्थानीय उच्चिष्ठ है

(B) $n_1 = 4, n_2 = 3$, के लिए ऐसे $\alpha \in (3, 5)$ का अस्तित्व है, जहाँ f स्थानीय निम्निष्ठ है

(C) $n_1 = 3, n_2 = 5$ के लिए ऐसे $\alpha \in (3, 5)$ का अस्तित्व है, जहाँ f स्थानीय उच्चिष्ठ है

(D) $n_1 = 4, n_2 = 6$ के लिए ऐसे $\alpha \in (3, 5)$ का अस्तित्व है, जहाँ f स्थानीय उच्चिष्ठ है

Ans. C

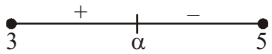
Sol. $f(3) = f(5) = 0$

$\to f'(\alpha) = 0 \qquad \alpha \in (3, 5)$ (apply Rolles)

$f'(x) = n_1(x-3)^{n_1-1}(x-5)^{x_2} + n_2(x-3)^{n_1}(x-5)^{x_2-1} = (x-3)^{n_1-1}(x-5)^{n_2-1}\left[n_1(x-5)+n_2(x-3)\right]$

$n_2 \in$ even

$n_2 - 1 \in$ odd



For $n_2 \in$ even $\qquad \alpha \to$ pt of maxima

$n_2 \in$ odd $\qquad \alpha \to$ pt of minima

Correct option is (C)

42. Let $f(x) = |(x-1)(x^2-2x-3)| + x - 3, x \in R$. If m and M are respectively the number of points of local minimum and local maximum of f in the interval (0, 4), then m + M is equal to _____.

[JEE(Main)June2022]

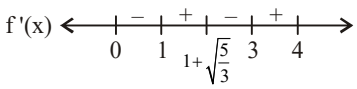
माना $f(x) = |(x-1)(x^2-2x-3)| + x - 3, x \in R$ है। यदि अंतराल (0, 4) में f के स्थानीय निम्नतम तथा स्थानीय उच्चतम बिन्दुओं की संख्या क्रमशः m + M हैं, तो m + M बराबर है _____ **[JEE(Main)June2022]**

Ans. 3

Sol. $f(x) = |(x-1)(x-3)(x+1)| + (x-3)$

$$f(x) = \begin{cases} x^3 - 3x^2 & , \quad x \in (0,1) \\ -x^3 + 3x^2 + 2x - 6 & , \quad x \in (1,3) \\ x^3 - 3x^2 & , \quad x \in (3,4) \end{cases}$$

$$f'(x) = \begin{cases} 3x^2 - 6x & , \quad x \in (0,1) \\ -3x^2 + 6x + 2 & , \quad x \in (1,3) \\ 3x^2 - 6x & , \quad x \in (3,4) \end{cases}$$



local minima at $x = 1, 3$

local maxima at $x = 1+\sqrt{\frac{5}{3}}$

$m = 2, M = 1$

$m + M = 3$

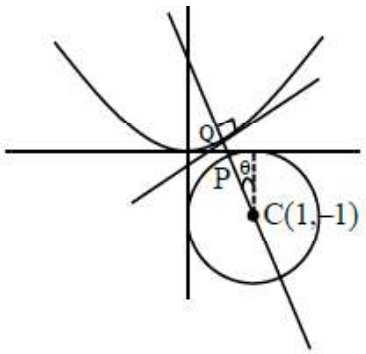
43. Let P and Q be any points on the curves $(x-1)^2+(y+1)^2=1$ and $y=x^2$, respectively. The distance between P and Q is minimum for some value of the abscissa of P in the interval : **[JEE(Main)July2022]**

माना वक्रों $(x-1)^2+(y+1)^2=1$ तथा $y=x^2$ पर क्रमशः P तथा Q कोई भी बिन्दु हैं। यदि P तथा Q के बीच दूरी निम्नतम है, तो P के भुजा का मान किस अंतराल में है : **[JEE(Main)July2022]**

(A) $\left(0,\frac{1}{4}\right)$ (B) $\left(\frac{1}{2},\frac{3}{4}\right)$ (C) $\left(\frac{1}{4},\frac{1}{2}\right)$ (D) $\left(\frac{3}{4},1\right)$

Ans. C

Sol.



$Q = (t, t^2)$

$m_{CQ} = m_{normal}$

$$\frac{t^2+1}{t-1} = -\frac{1}{2t}$$

Let $f(t) = 2t^3+3t-1$

$$f\left(\frac{1}{4}\right)f\left(\frac{1}{3}\right)<0 \Rightarrow t\in\left(\frac{1}{4},\frac{1}{3}\right)$$

$P \equiv (1+\cos(90+\theta), -1+\sin(90+\theta))$

$P = (1-\sin\theta, -1+\cos\theta)$

$$m_{normal} = m_{CP} \Rightarrow -\frac{1}{2t} = \frac{\cos\theta}{-\sin\theta} \Rightarrow \tan\theta = 2t$$

$$x = 1-\sin\theta = 1-\frac{2t}{\sqrt{1+4t^2}} = g(t)(let)$$

$$\Rightarrow g'(t) < 0$$

$g(t) \downarrow$ function

$t \in \left(\frac{1}{4}, \frac{1}{3}\right)$

$\Rightarrow g(t) \in (0.44, 0.485) \in \left(\frac{1}{4}, \frac{1}{2}\right)$

44. Let the area enclosed by the x-axis, and the tangent and normal drawn to the curve $4x^3 - 3xy^2 + 6x^2 - 5xy - 8y^2 + 9x + 14 = 0$ at the point (–2, 3) be A. Then 8A is equal to _____. **[JEE(Main)July2022]**

माना वक्र $4x^3 - 3xy^2 + 6x^2 - 5xy - 8y^2 + 9x + 14 = 0$ के बिन्दु (–2, 3) पर खींची गई स्पर्ष रेखा तथा अभिलंब और x-अक्ष से घिरे क्षेत्र का क्षेत्रफल A है। तो 8A बराबर है _____ **[JEE(Main)July2022]**

Ans. 170

Sol. $4x^3 - 3xy^2 + 6x^2 - 5xy - 8y^2 + 9x + 14 = 0$

differentiating both sides we get

$12x^2 - 3y^2 - 6xyy' + 12x - 5y - 5xy' - 16yy' + 9 = 0$

$\downarrow (-2, 3)$

$\Rightarrow 48 - 27 + 36y' - 24 - 15 + 10y' - 48y' + 9 = 0$

$\Rightarrow 29' = -y$

$\Rightarrow m_T = \frac{-9}{2}$ & $m_N = \frac{2}{9}$

$T \equiv y - 3 = \frac{-9}{2}(x + 2)$ & $N \equiv y - 3 = \frac{2}{9}(x + 2)$

$\downarrow y = 0$ $\downarrow y = 0$

$x = \frac{-4}{3}$ $x = \frac{-31}{2}$

$\therefore$ Area $= \frac{1}{2} \times \text{Base} \times \text{Height}$

$= 8A = 170$

45. Let M and N be the number of points on the $y^5 - 9xy + 2x = 0$, where the tangents to the curve are parallel to x-axis and y-axis, respectively. Then the value of M + N equals _____. **[JEE(Main)July2022]**

माना वक्र $y^5 - 9xy + 2x = 0$ पर उन बिन्दुओं, जिन पर स्पर्श रेखा x - अक्ष y-अक्ष के समान्तर हैं, कि संख्या क्रमशः M तथा N है। तो M + N बराबर है _____. **[JEE(Main)July2022]**

Ans. 2

Sol. $y^5 - 9xy + 2x = 0$

$$5y^4 \frac{dy}{x} - 9x\frac{dy}{dx} - 9y + 2 = 0$$

$$\frac{dy}{dx}\left(5y^4 - 9x\right) = 9y - 2$$

$$\frac{dy}{dx} = \frac{9y-2}{5y^4 - 9x} = 0 \text{ (for horizontal tangent)}$$

$y = \frac{2}{9} \Rightarrow$ Which does not staisfy the original equation $\Rightarrow M = 0$.

Now $5y^4 - 9x = 0$ (for vertical tangent)

$5y^4 (9y - 2) - 9y5 = 0$

$y^4[45y - 10 - 9y] = 0$

$y = 0$ (Or) $36y = 10$

$$y = \frac{5}{18}$$

$$y = 0 \Rightarrow x = 0 \;\&\; y = \frac{5}{18} \Rightarrow x =$$

(0, 0) $\left(x, \frac{5}{18}\right)$

$N = 2$

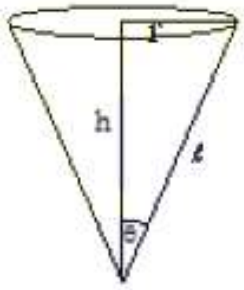
$M + N = 0 + 2 = 2$

46. A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1}\frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____. **[JEE(Main)July2022]**

एक पानी की टंकी लंब वृत्तीय शंकु, जिसका अक्ष ऊर्ध्वाधर तथा शीर्ष नीचे की ओर है, के आकार की है। इसका अर्ध शीर्ष कोण $\tan^{-1}\frac{3}{4}$ है। इसमें 6 घन मीटर प्रति घंटे की दर से पानी डाला जाता है। जब टंकी में पानी की गहराई 4 मीटर है, उस समय टंकी के गीले वक्र पृष्ठीय क्षेत्रफल के बढ़ने की दर (वर्ग मीटर प्रति घंटे में) है _____। **[JEE(Main)July2022]**

Ans. 5

Sol.



$$\tan\theta = \frac{3}{4} = \frac{r}{h}$$

$$\frac{dV}{dt} = 6$$

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi h^3 \tan^2\theta = \frac{9\pi}{48}h^3 = \frac{3\pi}{16}h^3$$

$$\Rightarrow \frac{dV}{dt} = \frac{3\pi}{16}.3h^2.\frac{dh}{dt} = 6 \Rightarrow \left(\frac{dh}{dt}\right)_{h=4} = \frac{2}{3\pi} m/hr$$

Now, $S = \pi r\ell = \frac{15}{16}\pi h^2$

$$\Rightarrow \frac{dS}{dt} = \frac{15\pi}{16}.2h\frac{dh}{dt}$$

$$\Rightarrow \left(\frac{dS}{dt}\right)_{h=4} = 5m^3/hr$$

47. If the tangent to the curve $y = x^3 - x^2 + x$ at the point (a, b) is also tangent to the curve $y = 5x^2 + 2x - 25$ at the point (2, –1), then $|2a + 9b|$ is equal to _____. **[JEE(Main)July2022]**

यदि वक्र $y = x^3 - x^2 + x$ के बिन्दु (a, b) पर स्पर्श रेखा, वक्र $y = 5x^2 + 2x - 25$ की भी बिन्दु (2, –1) पर स्पर्श रेखा है, तो $|2a + 9b|$ बराबर _____है। **[JEE(Main)July2022]**

Ans. 195

Sol. $y = 5x^2 + 2x - 25$ P(2, –1)

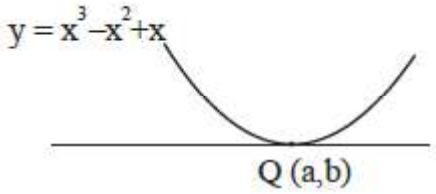
$y' = 10x + 2$

$y'_P = 22$

$\therefore$ tangent to curve at P

$y + 1 = 22(x - 2)$

$y = 22x - 45$



$\left.\frac{dy}{dx}\right|_{C_2} = 3x^2 - 2x + 1$

$\left.\frac{dy}{dx}\right|_{Q} = 3a^2 - 2a + 1$

Hence $3a^2 - 2a + 1 = 22$

$\therefore\ 3a^2 - 2a - 21 = 0$

$3a^2 - 9a + 7a - 21 = 0$

$(3a + 7)(a - 3) = 0 \Rightarrow \begin{matrix} a = 3 \\ a = -7/3 \end{matrix}$

from curve $b = a^3 - a^2 + a$

$a = 3$

$b = 21$ $\qquad |2a + 9b| = 195$

at $a = -7/3$ tangent will be parallel

Hence it is rejected

48. If the maximum value of a, for which the function $f_a(x) = \tan^{-1} 2x - 3ax + 7$ is non-decreasing in $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$, is $\bar{a}$, then $f_{\bar{a}}\left(\frac{\pi}{8}\right)$ is equal to : **[JEE(Main)July2022]**

यदि a का अधिकतम मान, जिसके लिए फलन $f_a(x) = \tan^{-1} 2x - 3ax + 7$, अंतराल $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$ में ह्रासमान नहीं है, $\bar{a}$ है तो $f_{\bar{a}}\left(\frac{\pi}{8}\right)$ बराबर है : **[JEE(Main)July2022]**

(A) $8 - \frac{9\pi}{4\left(9 + \pi^2\right)}$ (B) $8 - \frac{4\pi}{9\left(4 + \pi^2\right)}$

(C) $8\left(\frac{1 + \pi^2}{9 + \pi^2}\right)$ (D) $8 - \frac{\pi}{4}$

Ans. A

Sol. $f(x) = \tan^{-1} 2x - 3ax + 7$

$f'_a(x) = \frac{2}{1 + 4x^2} - 3a \geq 0$

$$a \le \left(\frac{2}{3\left(1+4x^2\right)}\right)_{\text{min.}} \quad \text{at } x = \pm\frac{\pi}{6}$$

$$a_{max} = \bar{a} = \frac{6}{9+\pi^2}$$

$$f_{\bar{a}}\left(\frac{\pi}{8}\right) = \tan^{-1}\frac{\pi}{4} - 3\frac{6}{9+\pi^2 8} + 7 = \tan^{-1}\frac{\pi}{4} - \frac{9\pi}{4\left(\pi^2+9\right)} + 7$$

49. Let a function $f: R \to R$ be defined as :

$$f(x) = \begin{cases} \int_0^x \left(5 - |t-3|\right)dt & , \quad x > 4 \\ x^2 + bx & , \quad x \le 4 \end{cases}$$

where be $b \in R$. If f is continuous at $x = 4$, then which of the following statements is NOT true ?

[JEE(Main)July2022]

(A) f is not differentiable at $x = 4$

(B) $f'(3) + f'(5) = \frac{35}{4}$

(C) f is increasing in $\left(-\infty, \frac{1}{8}\right) \cup (8, \infty)$

(D) f has a local minima at $x = \frac{1}{8}$

माना एक फलन $f: R \to R$

$$f(x) = \begin{cases} \int_0^x \left(5 - |t-3|\right)dt & , \quad x > 4 \\ x^2 + bx & , \quad x \le 4 \end{cases}$$

जहाँ $b \in R$ है, द्वारा परिभाषित है। यदि $x = 4$ पर f संतत है, तो निम्न में से कौनसा कथन सत्य नहीं है ?

[JEE(Main)July2022]

(A) $x = 4$ पर f अवकलनीय नहीं है

(B) $f'(3)+f'(5)=\frac{35}{4}$ है

(C) $\left(-\infty,\frac{1}{8}\right)\cup(8,\infty)$ में f वर्धमान है

(D) $x=\frac{1}{8}$, f का एक स्थानीय निम्नतम बिन्दु है

Ans. C

Sol. Given $f(x)\begin{cases}\int_0^x(5-|t-3|)dt, & x>4\\ x^2+bx, & x\le 4\end{cases}$

f(x) is continuousat $x=4$

So $\lim\limits_{x\to4^-}f(x)=\lim\limits_{x\to4^+}f(x)=f(4)$

So $16+4b=\int_0^3(2-t)dt+\int_3^4(8-t)dt$

$\Rightarrow 16+4b=15$

So $b=\frac{-1}{4}$

At $x=4$

$LHD=2x+b=\frac{31}{4}$

$RHD=5-|x-3|=4$

$LHD\neq RHD$

Option (A) is true

and $f'(3)+f'(5)=\frac{23}{4}+3=\frac{35}{4}$

Option (B) is true

$\because f(x)=x^2-\frac{x}{4}$ at $x\le 4$

$f'(x)=2x-\frac{1}{4}$

This function is not increasing.

In the interval in $x\in\left(-\infty,\frac{1}{8}\right)$

Option (C) is NOT TRUE.

This function f(x) is also local minima at $x = \frac{1}{8}$

50. The function $f(x) = xe^{x(1-x)}, x \in R$, is : **[JEE(Main)July2022]**

(A) increasing in $\left(-\frac{1}{2}, 1\right)$ (B) decreasing in $\left(\frac{1}{2}, 2\right)$

(C) increasing in $\left(-1, -\frac{1}{2}\right)$ (D) decreasing in $\left(-\frac{1}{2}, \frac{1}{2}\right)$

फलन $f(x) = xe^{x(1-x)}, x \in R$: **[JEE(Main)July2022]**

(A) $\left(-\frac{1}{2}, 1\right)$ में वर्धमान है (B) $\left(\frac{1}{2}, 2\right)$ में ह्यसमान है

(C) $\left(-1, -\frac{1}{2}\right)$ में वर्धमान है (D) $\left(-\frac{1}{2}, \frac{1}{2}\right)$ में ह्यसमान है

Ans. A

Sol. $f(x) = x\, e^{x(1-x)}$

$f'(x) = -e^{x(1-x)}(2x + 1)(x - 1)$

f(x) is increasing in $\left(-\frac{1}{2}, 1\right)$

51. The number of distinct real roots of the equation $x^5(x^3 - x^2 - x + 1) + x(3x^3 - 4x^2 - 2x + 4) - 1 = 0$ is _____. **[JEE(Main)July2022]**

$x^5(x^3 - x^2 - x + 1) + x(3x^3 - 4x^2 - 2x + 4) - 1 = 0$ के भिन्न वास्तविक मूलों की संख्या है _____ **[JEE(Main)July2022]**

Ans. 3

Sol. $x^5(x^3 - x^2 - x + 1) + x(3x^3 - 4x^2 - 2x + 4) - 1 = 0$

$\Rightarrow (x-1)^2 (x + 1)(x^5 + 3x - 1) = 0$

Let $f(x) = x^5 + 3x - 1$

$f'(x) > 0 \ \forall \ x \in R$

Hence 3 real distinct roots.

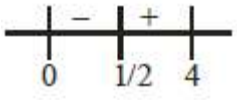
52. Let the function $f(x) = 2x^2 - \log_e x$, $x > 0$, be decreasing in (0, a) and increasing in (a, 4). A tangent to the parabola $y^2 = 4ax$ at a point P on it passes through the point (8a, 8a – 1) but does not pass through the point $\left(-\frac{1}{a}, 0\right)$. If the

equation of the normal at P is $\frac{x}{\alpha}+\frac{y}{\beta}=1$, then $\alpha+\beta$ is equal to _____. **[JEE(Main)July2022]**

फलन $f(x) = 2x^2 - \log_e x$, $x > 0$, अंतराल $(0, a)$ में ह्रासमान है तथा $(a, 4)$ में वर्धमान है। $y^2 = 4ax$ के एक बिन्दु P पर स्पर्श रेखा, बिन्दु $(8a, 8a-1)$ से होकर जाती है, परन्तु बिन्दु $\left(-\frac{1}{a}, 0\right)$ से होकर नहीं जाती है। यदि बिन्दु P अभिलंब का समीकरण $\frac{x}{\alpha}+\frac{y}{\beta}=1$ है, तो $\alpha + \beta$ बराबर है। **[JEE(Main)July2022]**

Ans. 45

Sol.



$$f'(x) = 4x - \frac{1}{x}$$

$$a = \frac{1}{2}$$

Let $P(x_1, y_1)$ be any point on $y^2 = 4ax$

$$\frac{1}{y_1} = \frac{3-y_1}{4-x_1} \Rightarrow y_1^2 - 6y_1 + 8 = 0$$

$y_1 = 2, 4$

$\Rightarrow$ P(8, 4) as P(2, 2) rejected

Equation of normal at P.

$y - 4 = -4(x - 8)$

$$\frac{x}{9}+\frac{y}{36}=1$$

$\alpha = 9$, $\beta = 36$

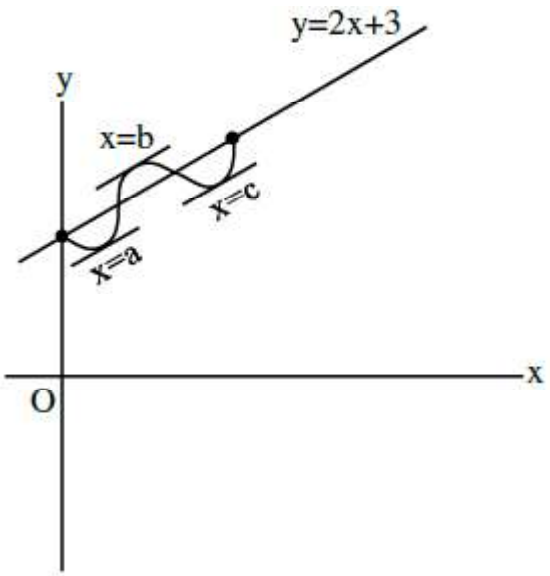
$\alpha + \beta = 45$

53. Let $f : [0, 1] \to R$ be a twice differentiable function in (0, 1) such that f(0) = 3 and f(1) = 5. If the line $y = 2x + 3$ intersects the graph of f at only two distinct points in (0, 1), then the least number of points $x \in (0, 1)$, at which $f''(x) = 0$, is _____. **[JEE(Main)July2022]**

माना $f : [0, 1] \to R$, अंतराल (0, 1) में दो बार अवकलनीय है तथा f(0) = 3 हैं। यदि रेखा f(1) = 5 हैं। यदि रेखा $y = 2x + 3$, f के ग्राफ को (0, 1) में केवल दो भिन्न बिन्दुओं पर काटती है, तो बिन्दुओं $x \in (0, 1)$ की न्यूनतम संख्या, जिन पर $f''(x) = 0$ है, है_____ **[JEE(Main)July2022]**

Ans. 2

Sol.



$f'(a) = f'(b) = f'(c) = 2$

$\Rightarrow f''(x)$ is zero

for atleast $x_1 \in (a, b)$ & $x_2 \in (b, c)$

54. If the absolute maximum value of the function $f(x) = (x^2 - 2x + 7)\ e^{\left(4x^3 - 12x^2 - 180x + 31\right)}$ in the interval $[-3, 0]$ is $f(\alpha)$, then : **[JEE(Main)July2022]**

यदि अंतराल $[-3, 0]$ में फलन $f(x) = (x^2 - 2x + 7)\ e^{\left(4x^3 - 12x^2 - 180x + 31\right)}$ का निरपेक्ष उच्चतम मान $f(\alpha)$ है, तो : **[JEE(Main)July2022]**

(A) $\alpha = 0$ (B) $\alpha = -3$ (C) $\alpha \in (-1, 0)$ (D) $\alpha \in (-3, -1]$

Ans. B

Sol. $f'(x) = e^{\left(4x^3 - 12x^2 - 180x + 31\right)}\left(12\left(x^2 - 2x + 7\right)(x + 3)(x - 5) + 2(x - 1)\right)$

for $x \in [-3, 0]$

$\Rightarrow f'(x) < 0$

$f(x)$ is decreasing function on $[-3, 0]$

The absolute maximum value of the function $f(x)$ is at $x = -3$

$\Rightarrow \alpha = -3$

55. The curve $y(x) = ax^3 + bx^2 + cx + 5$ touches the x-axis at the point $P(-2, 0)$ and cuts the y-axis at the point Q, where y′ is equal to 3. Then the local maximum value of $y(x)$ is : **[JEE(Main)July2022]**

वक्र $y(x) = ax^3 + bx^2 + cx + 5$, x-अक्ष को बिन्दु $P(-2, 0)$ पर स्पर्ष करता है तथा y-अक्ष को बिन्दु Q पर y′ का मान 3 है। तो $y(x)$ का स्थानीय उच्चतम मान है : **[JEE(Main)July2022]**

(A) $\frac{27}{4}$ (B) $\frac{29}{4}$ (C) $\frac{37}{4}$ (D) $\frac{9}{2}$

Ans. A

Sol. $y(x) = ax^3 + bx^2 + cx + 5$ is passing through

(-2,0) then $8a - 4b + 2c = 5$.....(1)

$y'(x) = 3ax^2 + 2bx + c$ touches x-axis at (-2,0)

$12a - 4b + c = 0$ .....(2)

again, for $x = 0$, $y'(x) = 3$

$c = 3$ .....(3)

Solving eq. (1), (2) & (3) $a = -\frac{1}{2}, b = -\frac{3}{4}$

$$y'(x) = -\frac{3}{2}x^2 - \frac{3}{2}x + 3$$

y(x) has local maxima at $x = 1$

$$y(1) = \frac{27}{4}$$

56. Let $f(x) = \begin{cases} x^3 - x^2 + 10x - 7, & x \le 1 \\ -2x + \log_2\left(b^2 - 4\right), & x > 1 \end{cases}$. Then the set of all values of b, for which f(x) has maximum value at $x = 1$, is : **[JEE(Main)July2022]**

माना $f(x) = \begin{cases} x^3 - x^2 + 10x - 7, & x \le 1 \\ -2x + \log_2\left(b^2 - 4\right), & x > 1 \end{cases}$ है। तो b के सभी मानों, जिनके लिए f(x) का अधिकतम मान $x = 1$ पर है, का समुच्चय है : **[JEE(Main)July2022]**

(A) $(-6, -2)$ (B) $(2, 6)$ (C) $[-6, -2) \cup (2, 6]$ (D) $\left[-\sqrt{6}, -2\right) \cup \left(2, \sqrt{6}\right]$

Ans. C

Sol. $f(1) = 3$

For $x < 1$, $f'(x) = 3x^2 - 2x + 10 > 0$

$\Rightarrow$ f(x) is increasing

For $x > 1$, $f'(x) < 0$

$\Rightarrow$ function is decreasing.

$$\lim_{x \to 1^+} f(x) = -2 + \log_2\left(b^2 - 4\right)$$

For maximum value at $x = 1$

$$3 \ge -2 + \log_2\left(b^2 - 4\right)$$

$32 \geq b^2 - 4 > 0$

$b \in [-6,-2] \cup (2,6]$

57. The sum of the absolute maximum and absolute minimum values of the function $f(x) = \tan^{-1}(\sin x - \cos x)$ in the interval $[0, \pi]$ is : **[JEE(Main)July2022]**

अंतराल $[0, \pi]$ में, फलन $f(x) = \tan^{-1}(\sin x - \cos x)$ के निरपेक्ष उच्चतम तथा निरपेक्ष निम्न मानों का योग है :

[JEE(Main)July2022]

(A) 0 (B) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) - \frac{\pi}{4}$ (C) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right) - \frac{\pi}{4}$ (D) $\frac{-\pi}{12}$

Ans. C

Sol. $f(x) = \tan^{-1}(\sin x - \cos x)$

$$f'(x) = \frac{\cos x + \sin x}{(\sin x - \cos x)^2 + 1} = 0$$

$$\therefore x = \frac{3\pi}{4}$$

x	0	$\frac{3\pi}{4}$	π
$f(x)$	$-\frac{\pi}{4}$	$\tan^{-1}\sqrt{2}$	$\frac{\pi}{4}$

$$\therefore \left.\begin{array}{l} (f(x))_{max} = \tan^{-1}\sqrt{2} \\ (f(x))_{min} = -\frac{\pi}{4} \end{array}\right]$$

$$\text{sum} = \tan^{-1}\sqrt{2} - \frac{\pi}{4}$$

$$= \cos^{-1}\frac{1}{\sqrt{3}} - \frac{\pi}{4}$$

58. Let $f(x) = 3^{(x^2-2)^3+4}$, $x \in R$. Then which of the following statements are true?

P : $x = 0$ is a point of local minima of f

Q : $x = \sqrt{2}$ is a point of inflection of f

R : f' is increasing for $x > \sqrt{2}$ **[JEE(Main)July2022]**

(A) Only P and Q (B) Only P and R

(C) Only Q and R (D) All P, Q and R

माना $f(x)=3^{(x^2-2)^3+4}$, $x \in R$ है। तो निम्न कथनों में से कौनसे सत्य है ?

P : $x = 0$, f का एक स्थानीय निम्ननिष्ठ बिन्दु है :

Q : $x=\sqrt{2}$,f का एक नति परिवर्तन बिन्दु है

R: $x > \sqrt{2}$ के लिए f' वर्धमान है **[JEE(Main)July2022]**

(A) केवल P तथा Q (B) केवल P तथा R

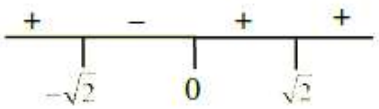
(C) केवल Q तथा R (D) सभी P, Q तथा R

Ans. D

Sol. $f(x)=81.3^{(x^2-2)^3}$

$f'(x)=81\cdot 3^{(x^2-2)^3}\cdot \ln 3\cdot 3(x^2-2)^2\cdot 2x$

$=(81\times 6)3^{(x^2-2)^3}\, x(x^2-2)^2 \ln 3$



$x = 6$ is point of local min

$f'(x)=\underbrace{(486\cdot \ln 3)}_{k}\underbrace{3^{(x^2-2)^3}\, x(x^2-2)^2}_{g(x)}$

$g'(x)=3^{(x^2-2)^3}(x^2-2)^2+x\cdot 3^{(x^2-2)^3}\cdot 4x\cdot(x^2-2)$

$+x\cdot(x^2-2)^2\cdot 3^{(x^2-2)^3}\ln 3\cdot 3(x^2-2)^2\cdot 2x$

$=3^{(x^2-2)^3}(x^2-2)\left[x^2-2+4x^2+6x^2\ln 3(x^2-2)^3\right]$

$g'(x)=3^{(x^2-2)^3}(x^2-2)\left[5x^2-2+6x^2\ell \ln 3(x^2-2)^3\right]$

$f''(x)=k\cdot g'(x)$

$f''(\sqrt{2})=0, f''(\sqrt{2}^+)>0, f''(\sqrt{2}^-)<0$

$x=\sqrt{2}$ is point of inflection

$f''(x) > 0$ for $x > \sqrt{2}$ so f' is increasing

59. The sum of the maximum and minimum values of the function $f(x) = |5x - 7| + [x^2 + 2x]$ in the interval $\left[\frac{5}{4}, 2\right]$, where [t] is the greatest integer $\le t$, is _____. **[JEE(Main)July2022]**

फलन $f(x) = |5x - 7| + [x^2 + 2x]$, जहाँ [t] महत्तम पूर्णांक $\le t$ है, के अन्तराल $\left[\frac{5}{4}, 2\right]$ में उच्चतम तथा निम्नतम मानों का योगफल है _____ **[JEE(Main)July2022]**

Ans. 15

Sol. $f(x) = [5x - 7] + [x^2 + 2x]$

$= [5x - 7] + [(x + 1)^2] - 1$

Critical points of

$$f(x) = \frac{7}{5}, \sqrt{5} - 1, \sqrt{6} - 1, \sqrt{7} - 1, \sqrt{8} - 1, 2$$

$\therefore$ Maximum or minimum value of f(X) occur at critical points or boundary points

$$\therefore \quad f\left(\frac{5}{4}\right) = \frac{3}{4} + 4 = \frac{19}{4}$$

$$f\left(\frac{7}{5}\right) = 0 + 4 = 4$$

as both $|5x - 7|$ and $x^2 + 2x$ are increasing in natuyre after $x = \frac{7}{5}$

$\therefore \quad f(2) = 3 + 8 = 11$

$\therefore \quad f\left(\frac{7}{5}\right)_{min} = 4$ and $f(2)_{max} = 11$

Sum is $4 + 11 = 15$